

Assignment 1 (Chapters 1 - 12)

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Exercise 1.1

$A = \{0, 1, 2, 3, 4, 5, 6, 7\}$, $B = \{0, 1, 2, 3, 4\}$, $C = \{2, 3, 5, 7\}$

a) $\{0, 1, 2, 3, 4, 5, 6, 7\}$ b) $\{0, 1, 2, 3, 4\}$ c) $\{5, 6, 7\}$ d) $\{\}$
e) $\{6\}$ f) $\{\}$ g) $\{\}$ h) $\{0, 1, 2, 3, 4\}$

Exercise 2.1

a) $E \rightarrow S$ b) $S \rightarrow E$ c) $E \wedge \neg S$ d) $(S \rightarrow \neg E) \wedge (\neg E \rightarrow S)$

Exercise 3.2

By looking at the truth tables I found the following relations:

1. $A \rightarrow B \Leftrightarrow \neg(\neg A \wedge B) \Leftrightarrow \neg A \vee B$
2. $\neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$

After that, I tried to simplify everything into just NOT, OR, and AND and see which ones are the same. Thus, I got the following result:

- a) $\neg P \vee (Q \wedge R)$
- b) $\neg P \vee \neg Q \vee R$
- c) $P \vee \neg Q \vee R$
- d) $\neg P \vee (Q \wedge R)$
- e) $\neg P \vee \neg Q \vee R$
- f) $P \vee \neg Q \vee R$

Although after e) I didn't need to find f) since from the hint we could deduce that it is guaranteed to be equivalent to the one left over. Now, as you can see, a) is equivalent to d), b) is equivalent to e), and c) is equivalent to f).

Exercise 4.3

a) No, we cannot deduce A . From $A \vee B$ we can deduce either A , or B , or $A \wedge B$, but there is no guarantee to deduce A since there exists a possibility, namely: B where A is not present. From $C \wedge B$ we can deduce both B and C , but B does not help much previous deduction; all it does is it eliminates the possibility of A alone, but $A \wedge B$ is still possible, although so is B alone. We also know that if A then C , but that does not mean that if C then A . Nowhere can we find a direct deduction for A , nor a deduction through other connections.

b) Yes. From $A \wedge C$ we know that both A and C must be true. We also know that if C then B , but also just B . Since we know that A and B , we can safely deduce $A \wedge B$.

Exercise 5.4

Let's assume that $(P \rightarrow (Q \rightarrow P))$ is not a tautology. This means that it is falsifiable. for it to be falsifiable there must exist an evaluation of $(P \rightarrow (Q \rightarrow P))$ where it is $\perp$. There exists only one such example where P is $\top$ and $(Q \rightarrow P)$ is $\perp$. However, to get $(Q \rightarrow P)$ to be false Q must be $\top$ and P must be $\perp$. But wait, P can not be $\perp$ because it must be $\top$ from the start. Uh-oh, a contradiction! Therefore, $(P \rightarrow (Q \rightarrow P))$ can not be false. this means that all the interpretations of $(P \rightarrow (Q \rightarrow P))$ evaluate to true which according to the definition is a tautology.

Exercise 6.2

The relation R is as follows:

$R = \{\langle 3, 1 \rangle, \langle 6, 1 \rangle, \langle 6, 3 \rangle, \langle 7, 1 \rangle, \langle 12, 1 \rangle, \langle 12, 3 \rangle, \langle 12, 6 \rangle, \langle 15, 1 \rangle, \langle 15, 3 \rangle, \langle 17, 1 \rangle\}$. Due to the rules x/y and "et naturlig tall større enn 1" there does not exist a relation which refers to itself. Therefore, R is irreflexive which also makes it not reflexive. R is also not symmetrical and thus anti-symmetrical. Finally, R is transitive.

Exercise 7.1

- a) g is not a well-defined function because it is missing an element 3 from S .
- b) h is a valid function. The function is not injective because the outputs (codomain values) are not unique. The function is also not surjective because it is missing codomain values for a , b , and c .
- c) i is also a valid function that is injective (all outputs are unique) but not surjective. In fact, none of the functions can be surjective because there are not enough inputs (domain elements). The number of domain elements is smaller than the number of codomain elements.
- d) j is not valid because 3 from S corresponds to two different outputs. Which one of these two is the function supposed to give as an output is not defined well enough.

Exercise 8.2

- a) If x is in $\overline{A \cup B}$ it means that x can not be in $A \cup B$ which means that both A and B can not have an x . Now, since both A and B can not have an x , both $\overline{A}$ and $\overline{B}$ must have an x . Therefore, $\overline{A} \cap \overline{B}$ also has an x .
- b) If x is in $\overline{A} \cap \overline{B}$ it means that x must be in both $\overline{A}$ and $\overline{B}$. This means that x can not be in both A and B . Since x can not be in both A and B , x can not be in $A \cup B$. Therefore, $\overline{A \cup B}$ must have an x .
- c) From a) we established that $\overline{A \cup B} \Rightarrow \overline{A} \cap \overline{B}$, and from b) we established that $\overline{A} \cap \overline{B} \Rightarrow \overline{A \cup B}$. This means that $\overline{A \cup B} \Leftrightarrow \overline{A} \cap \overline{B}$.

Exercise 8.3

To show that $\mathbb{N}$ has the same cardinality as $\mathcal{H}$ we can try to map each value of $\mathbb{N}$ to its corresponding value in $\mathcal{H}$. For that we must create a function, let's call it f which we define as follows: $f(n) = n/2$. Now, since n becomes $n/2$ there is no possible way that the function is not injective; each value generated is unique. This function is also surjective, no values are missed since by definition $\mathcal{H}$ consists of all the positive halves of numbers and $\mathbb{N}$ consists of all the positive integers which our function accounts for by halving each integer. Now, since the function is both injective and surjective it is also bijective. This means that each value in $\mathbb{N}$ can be paired to its corresponding value in $\mathcal{H}$ which answers our question: Yes, they have the same cardinality.

Exercise 9.1

a) The reflexive closure of R is $R \cup \{\langle b, b \rangle, \langle c, c \rangle, \langle d, d \rangle\}$.

b) To make it symmetrical we need to make it $R \cup \{\langle b, a \rangle, \langle d, b \rangle\}$. To make it transitive we must add an additional $R \cup \{\langle b, b \rangle, \langle d, d \rangle, \langle a, d \rangle, \langle d, a \rangle\}$. In the end here is the summary of what he have to add to make it symmetrical and transitive $R \cup \{\langle a, d \rangle, \langle b, a \rangle, \langle b, b \rangle, \langle d, a \rangle, \langle d, b \rangle, \langle d, d \rangle\}$.

Exercise 10.1

a) Lets call our function f . First we define base case which we get from $+1$, so $f(0) = 1$. Then we define our recursive case which we add another $+3$ which we get from $3n$ each time, so $f(n+1) = f(n) + 3$. Here is a list of results starting from 0 up to and including 7:

$$f(0) = 1$$

$$f(1) = f(0) + 3 = 1 + 3 = 4$$

$$f(2) = \dots = 7$$

$$f(3) = \dots = 10$$

$$f(4) = \dots = 13$$

$$f(5) = \dots = 16$$

$$f(6) = \dots = 19$$

$$f(7) = \dots = 22$$

I did not show the steps for all simply because it takes too long.

b) Again, we call our function f , but this time the base case is much simpler being $f(0) = 0$. The recursive case however can have many forms:

$$1. f(n+1) = (\sqrt{f(n)} + 1)^2.$$

$$2. f(n+1) = f(n) + 2n + 1$$

$$3. f(n+1) = f(n) + 2\sqrt{f(n)} + 1$$

Lets take the second one since it is the most simple out of these three. Here are the 8 first values starting from 0 up to and including 7:

$$f(0) = 0$$

$$f(1) = f(0) + 2(0) + 1 = (0) + 2(0) + 1 = 0 + 0 + 1 = 1$$

$$f(2) = \dots = 4$$

$$f(3) = \dots = 9$$

$$f(4) = \dots = 16$$

$$f(5) = \dots = 25$$

$$f(6) = \dots = 36$$

$$f(7) = \dots = 49$$

Most of the steps take way to many rewrites to show, but you get the idea.

Exercise 10.4

We can define the formula by accounting for the cases we are limiting ourselves to which are: The main logical operations $\neg, \wedge, \vee, \rightarrow$. We will call our function f . Since we don't care about premises $f(P) = 0$. Now we will separate stuff like this:

$$f(\neg F) = f(F)$$

$$f(F \wedge G) = f(F) + f(G) + 1$$

$$f(F \vee G) = f(F) + f(G)$$

$$f(F \rightarrow G) = f(F) + f(G)$$

As you can see all of them are pretty much the same except for $f(F \wedge G)$ where we add an extra 1. That is because our goal is to count them, and this is the way we do it. At the end we are left with premises only, and the only way we can increase our counter is only by separating $\wedge$ operator.

Exercise 11.2

a) The statement $0 + 2 + 4 + 6 + \dots + 2 \cdot n = n \cdot (n + 1)$ claims that a sum of n first even numbers can be represented as $n \cdot (n + 1)$. In math language this is also known as $\sum_{i=0}^n 2i = n \cdot (n + 1)$. The task is to determine whether the equivalence is true or false.

b) The base case starts at $n = 0$. So, according to the formula we should get $(0) \cdot (0 + 1) = 0 \cdot 1 = 0$. That is correct, 0 is indeed the sum of 1 first even numbers. Thus the base case is true.

Exercise 11.3

a) The induction hypothesis says that if the base case is proven to be true, and if when we assume that it is true for both the given formula and $n = k$ we show that it is true for $n = k + 1$, then we can conclude that the formula we were proving in fact holds true. So the induction hypothesis in our case is $0 + 2 + 4 + 6 + \dots + 2 \cdot n \stackrel{IH}{=} n \cdot (n + 1)$.

b) Right away we start by assuming that the formula holds for $n = k$ where k is any valid input number. This means that $0 + 2 + 4 + 6 + \dots + 2 \cdot k = k \cdot (k + 1)$. Now we attempt to prove that $0 + 2 + 4 + 6 + \dots + 2 \cdot k + 2 \cdot (k + 1) = (k + 1) \cdot ((k + 1) + 1)$.

Since we know, or rather assumed, that $0 + 2 + 4 + 6 + \dots + 2 \cdot k = k \cdot (k + 1)$ we can now replace the summation form with this formula, so we get $k \cdot (k + 1) + 2 \cdot (k + 1) = (k + 1) \cdot ((k + 1) + 1)$. Our task now is to prove that the part on the left is equivalent to the part on the right.

Let's simplify both sides as much as we can. Left side: $k \cdot (k + 1) + 2 \cdot (k + 1) = (k^2 + k) + (2k + 2) = k^2 + 3k + 2$. Right side: $(k + 1) \cdot ((k + 1) + 1) = (k + 1)^2 + (k + 1) = (k^2 + 2k + 1) + (k + 1) = k^2 + 3k + 2$. As you can see at the end both sides are the same, $k^2 + 3k + 2 = k^2 + 3k + 2$ is hard to argue against.

Now, since we know that the base case is true, we can use this as our foundation which we use to prove the next base case. We know that the next base case must also be true from our induction case proof. Simply replace k with our base case and no assumptions are needed. That next case we then use for the next next case, and so on. Thus, we have just proven that the statement is in fact true and it holds for all the cases.

Exercise 12.3

a) The statement claims that whenever we apply a function *FLIP* twice as such: $FLIP(FLIP(F))$ where F is used as a placeholder for any formula no matter the simplicity or complexity, then the end result will be equivalent to the F which was used as an input.

b) To prove this using the induction method we will have to cover the base case first. That is, the case that can not be broken down further:

1. $FLIP(FLIP(p)) = FLIP(p) = p$ which means that $FLIP(FLIP(p)) \equiv p$. As you can see the base case does in fact hold.

Exercise 12.4

a) The induction hypothesis says that if the base cases are proven to be true, and if when we assume that the property holds for the immediate subformulas of some formula F , we can then use this to show that it also holds for F itself. If that works, then we can conclude that the formula we were proving in fact holds true. That is because we proved the base case, whose form can be made into a subformula of any formula that is not the base case itself, and so on. So in our case, that would be some formula F whose subformulas are A and B , for which we assume that: $FLIP(FLIP(A)) \stackrel{IH}{\equiv} A$ and $FLIP(FLIP(B)) \stackrel{IH}{\equiv} B$.

b) All we have to do now is to show that it holds for the rest of the cases, namely: $F = \neg A$, $F = A \wedge B$, $F = A \vee B$, and $F = A \rightarrow B$:

1. $FLIP(FLIP(\neg A)) = FLIP(\neg FLIP(A)) = FLIP(\neg A) = \neg FLIP(A) = \neg A \equiv \neg A$
2. $FLIP(FLIP(A \wedge B)) = FLIP(FLIP(A) \wedge FLIP(B)) = FLIP(A \wedge B) = A \wedge B \equiv A \wedge B$
3. $FLIP(FLIP(A \vee B)) = FLIP(FLIP(A) \vee FLIP(B)) = FLIP(A \vee B) = A \vee B \equiv A \vee B$
4. $FLIP(FLIP(A \rightarrow B)) = FLIP(FLIP(A) \rightarrow FLIP(B)) = FLIP(A \rightarrow B) = A \rightarrow B \equiv A \rightarrow B$

And just like that we have shown that in all the cases the premise holds true. Thus, due to our proof of the base case and proof of recursive case through an assumption we can conclude that the premise holds.